AI-FIRST SDLC PLATFORM

Platform Overview, Phase-by-Phase Build, & Local Execution Guide

An engine-agnostic, multi-tenant, web-delivered agentic platform that runs the full software lifecycle — ideation through deployment — on a client's real tools of record, with a human review gate at every state change.

Prepared for	The Engineering Team — EdvenSoft
Scope	Phases 0–4.8 (complete & tested); Phase 5 next. Brownfield code-intelligence dispatch is a documented, deliberately deferred decision (see 2.8 and ADR references) — everything else is UI-driven end to end.
Document	Goal & overview · phase build descriptions · what works today end-to-end · hands-on local testing guide (UI-first) · setup & credentials
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1. What We Are Building — and What We Are Trying to Achieve

The product is not “an AI that writes code.” It is a platform that runs your entire software lifecycle the way a disciplined engineering organisation does — with specialists, evidence, review gates, and an audit trail — but with AI agents doing the heavy lifting and humans staying in control of every decision that matters.

1.1 The ultimate goal

We already had a powerful asset: a suite of **17 expert AI agents** (ideation, requirements, PRD, architecture, UX, task-breakdown, implementation, code review, testing, security, CI/CD, deployment, and more) plus the rules and gates that made them behave like senior engineers. The catch: that suite ran *locally*, glued to one developer's machine and Git checkout, with project context living in Markdown files.

The goal of this platform is to transform that suite into a **multi-tenant, web-delivered, engine-agnostic agentic SDLC platform** that:

- operates each client's **real tools of record** — Confluence, Jira, Figma, Xray, GitHub — bidirectionally, instead of inventing parallel copies;
- runs the **full lifecycle** (idea -> requirements -> PRD -> architecture -> UX -> build -> review -> test -> security -> CI/CD -> deploy) for both new (greenfield) and existing (brownfield) projects, single- or multi-repo;
- keeps a **human review gate** at every state change — nothing reaches a system of record without an approval;
- is **durable** (a run survives crashes, restarts, and an offline laptop) and **reproducible** (every run records the agent version, model, inputs, outputs, cost, and approver);
- treats the AI **engine as swappable** — Claude today, others later — without rewriting a single agent.

> **In one sentence**
>
> Take the intelligence of 17 senior-engineer AI agents, lift it off the laptop into a governed cloud-ready platform, point it at the client's real tools, and let teams ship software faster without giving up control, traceability, or quality.

1.2 The seven non-negotiables (the rules the whole system obeys)

Every design decision serves these locked principles. They are worth internalising because they explain *why* the platform is shaped the way it is.

#	Principle	What it means in plain terms
1	State vs. content	The platform owns **state** (pointers, provenance, events in Postgres). External tools own **content** (the actual Jira issue, Confluence page, Git PR). Local files are throwaway.
2	Orchestration vs. execution	The platform decides *what* happens and *when* (and runs the gates). The AI engine only *executes* a step. So engines stay interchangeable.
3	Memory = index, not copies	We store pointers + hashes + versions of artifacts, never copied bodies. This is what lets us detect drift and cross-check conflicts.

#	Principle	What it means in plain terms
4	Work is a graph	The unit of work is a **WorkItem**; work is a per-item DAG, not a rigid linear pipeline — features can progress independently.
5	Every artifact is gated	draft -> reviewed -> gated -> published. No auto-publish, ever. Nothing reaches a system of record before an approval.
6	Tenant isolation by RLS	Every database table carries a tenant_id and a Postgres Row-Level-Security policy from its first migration. One client can never see another's rows.
7	Capabilities, not vendors	Business logic calls tasktracker.* / docstore.* / codehost.* — never jira.* directly — so a connector can be swapped by editing one row.

Three more locked decisions round these out: the default engine is **Claude (Agent SDK)** behind an ExecutionDriver interface; durability is provided by **Temporal** from day one; and the agents are migrated by the **75 / 15 / 10** rule (reuse ~75% of an agent's reasoning verbatim, rewrite ~15% of its data references to the platform, replace ~10% of invocation glue).

1.3 The architecture at a glance — seven planes

The platform is organised into seven cooperating layers. You will meet each of these in the phase descriptions and again in the testing guide.

Plane	Responsibility	Where it lives
Experience	What people see and click: project workspace, phase console, gate approvals, traceability viewer.	React app (frontend/)
Control	The brain: Work-DAG state machine, Gate/Approval service, run scheduler, Manifest Composer, reconciliation, driver router.	Django services (backend/)
Execution	Runs the agents: the ExecutionDriver interface + ClaudeDriver; planning runs server-side, build runs on a developer-machine companion.	drivers/, runner/
Integration	Talks to the outside world through capability interfaces + connectors + the Platform-State MCP server + webhooks + a rate governor.	backend/integrations/, mcp_platform_state/
Memory / State	Postgres (with pgvector) + an S3-compatible object store: project profile, artifact registry, traceability graph, work DAG, event ledger, run history.	Postgres + MinIO
Security & Governance	Tenant isolation (RLS), prompt-injection defences, secret scanning, egress control, audit. (Deepens in Phase 5.)	across the stack
Observability & Quality	Durable workflow visibility, run cost metering, and the test/quality plane. (Deepens in Phase 5.)	Temporal + apps/quality/

1.4 What “done” looks like at each milestone

The build proceeds phase by phase, each ending in a working **thin vertical slice** demonstrated end-to-end. The headline milestone:

First usable product — after Phase 2

A user can take a **greenfield** project from idea -> PRD -> architecture -> backlog in Jira -> an implemented GitHub PR, and can **onboard an existing single-repo** project (from local documents or live Atlassian). Phases 3–4 add design/test/security depth and brownfield-full/multi-repo; Phases 5–6 add enterprise hardening, the cloud runner fleet, and engine choice.

2. The Build, Phase by Phase (Phases 0–4.8)

Each phase below lists the concrete components that were built and explains what each one does in everyday language, and closes with a **capstone slice** — the thin end-to-end demo that proves the phase works. (For a real, straight-through test pass — see the scenarios in **Section 5**.) Phases 0 through 4.8 are complete and tested. Phases 0–4 (2.1–2.5) built the platform's execution spine and connectors; Phases 4.5/4.7/4.8 (2.6–2.8) turned it into a real, self-serve, browser-driven product — every screen in this guide from Section 3 onward is one of the hi-fi screens 4.8 shipped.

2.1 Phase 0 — The Platform Spine

Goal: move the source of truth off the filesystem and run the existing agents through a real platform with state, gates, and provenance — before adding any external tools. This phase builds the skeleton everything else hangs on.

Component	What it is / does
State plane (Postgres)	The durable memory: Tenant, Project, **Project Profile** (the database record that replaces the old CLAUDE.md), Repository, **WorkItem** (the unit of work + its DAG), **Artifact Registry** (pointers + hashes + versions, never bodies), **Event Ledger** (append-only history), **Run History**, AgentSpec Registry.
Platform-State MCP server	A small tool server the agents call to read/write platform state: get_context, register_artifact, log_event, add_trace_edge, list/transition_work_item, etc. Every call is tenant-scoped.
Manifest Composer	Assembles a fresh, budget-bounded AGENTS.md per run from the Project Profile + relevant artifact pointers + recent events + rules. This is the context an agent reads — generated each run, never hand-maintained.
Temporal AgentRunWorkflow	The durable orchestration of one agent run: compose context -> execute agent -> verify it wrote what it should -> open the gate -> wait for a human -> publish + advance. Survives restarts; retries are idempotent.
Gate / Approval service	Gates are control-plane Approval objects, never engine hooks. A human with the right role decides; the decision signals the workflow to resume. (Phase 3 added WCAG checklist conditions to this.)
ExecutionDriver + ClaudeDriver	The seam that makes the engine swappable. Business logic calls the interface; the ClaudeDriver runs the Claude Agent SDK underneath.
Tenant isolation (RLS)	A non-superuser database role the app connects as, plus a Row-Level-Security policy on every table. Unset tenant context fails closed (sees no rows).
React Experience plane	The web UI: create a project, edit its profile, a Phase Console to trigger phases and watch the run timeline, a gate-approval screen, an artifact view, and a traceability view.

Component	What it is / does
Migrated agents 01–04	Ideation, Requirements, PRD, Architecture — the first agents migrated by the 75/15/10 rule, plus the rules and per-agent triggers that replace the old slash commands.

> **Phase 0 — capstone slice (what this phase proves)**
>
> Run ideation -> requirements -> PRD -> architecture fully on platform state; artifacts registered as drafts in object storage; every step gated by a human; runs attributed; the manifest rendered per run. (This is exactly what you will click through in the testing guide, Section 3.)

2.2 Phase 1 — First System of Record (Atlassian)

Goal: make the platform operate a client's real Confluence and Jira, bidirectionally, with progress tracking and traceability — and let existing projects be onboarded.

Component	What it is / does
DocStore connector (Confluence)	Creates/updates Confluence pages (versioned, read-through). The platform publishes a PRD page only after its gate is approved.
TaskTracker connector (Jira)	Creates epics/stories/subtasks, transitions issues, comments, searches — headless via API token.
Capability abstraction	docstore.* and tasktracker.* interfaces + a per-tenant get_capability() selector. Callers never name Jira or Confluence directly (non-negotiable #7).
Artifact publish path	The gate -> publish -> pointer+version lifecycle to the live SoR; idempotent; falls back to “internal” when no connector is configured.
Traceability graph + viewer	requirement -> PRD -> epic -> story lineage, with deep-links into the live Confluence/Jira items.
Reconciliation service	Turns proposed work into accepted SoR issues without ever duplicating something that already exists (register-don't-recreate).
Brownfield-lite onboarding	Adopt an existing project: register what is already in Atlassian, and extract structured artifacts from static **Word/PDF/Excel** documents (with provenance + a review gate).
Rate governor	A per-(tenant, capability) token bucket + backoff + short-lived read cache, so live API calls stay within limits and don't flake.
Migrated agent 06	Task Breakdown — decomposes the PRD/architecture into a vertically-sliced backlog that reconciliation pushes into Jira on approval.

2.3 Phase 2 — Build Execution (real code, real PRs, durably)

Goal: produce real code — implementation, review, test — as GitHub pull requests, durably, *without changing how a developer works day to day*.

Component	What it is / does
CodeHost connector (GitHub)	Owns the PR / branch / status / comment / merge surface via the GitHub REST API. The engine never calls Git directly; the platform does, behind the codehost.* capability.

Component	What it is / does
Local-companion runner	A small daemon a developer runs on **their own machine**. It enrolls with the platform, pulls a build task, and works in a clean, platform-managed **git worktree** (a feature/<key> branch) so the developer can run and test the change locally before any PR. Their main checkout is never touched.
Remote Platform-State serving	The companion uses the state tools over an authenticated remote transport — the dev machine needs no database credentials; tenant binding and RLS stay server-side.
run_build_task	The build-mode driver call: launches the agent in the worktree with file + shell tools, under a per-phase tool allowlist (no CodeHost, no arbitrary network).
Temporal BuildWorkflow (saga)	Durable build orchestration: dispatch to the companion -> await its report (or park for human takeover if the laptop is offline) -> open a draft PR -> review -> remediation loop -> reviewer merge gate -> merge. Idempotent; compensating on failure.
07 <-> 08 remediation loop	Implementation (07) and Code Review (08) cooperate: each blocking finding becomes a child WorkItem; 07 re-runs in remediation mode until 08 is clean — then the merge gate can be granted.
Injection guardrails v1	Secret-scan before push (blocks credentials reaching GitHub), an egress allowlist (only approved hosts), and the per-phase tool allowlist.
Human takeover + webhook reconcile	A developer can take a WorkItem over and just work by hand; a signed GitHub webhook reconciles their commits/PR/CI status back into the registry (register-don't-recreate).
Light multi-repo	One project can target several repositories (one repo per WorkItem); each builds on its own branch/PR independently.
Migrated agents 07, 08, 11, 12	Implementation, Code Review, CI/CD, Deployment — with their reasoning (Clean Code/TDD, the eleven review dimensions, the Jenkins/Actions templates, blue-green/canary releases) preserved verbatim.

Phase 2 — capstone slice (the stop-and-dogfood MVP)

WorkItem -> companion worktree -> local build/test -> PR -> CI status -> review -> remediation -> reviewer merge gate -> merge, with a forced mid-run failure recovered idempotently, a human commit reconciled via webhook, and two repos in one project building independently.

2.4 Phase 3 — Design & Test Loops

Goal: close the UX and QA loops — read a client's live design, push tests and results to Xray, and turn a failing test into a tracked, linked bug.

Component	What it is / does
DesignSource connector (Figma)	**Read-as-SoR**: the platform reads the client's *current* Figma design at run time (frames, design tokens, components) rather than regenerating it. Canvas write-back is a deferred seam.
Design-System / tokens + Code Connect	Aggregates the current design into a design_system artifact (tokens) and a code_connect mapping that links design components to code components.
a11y (WCAG) gate	The accessibility review is enforced as a **checklist of WCAG 2.1 AA conditions on the design Approval** — an approve is blocked until every required criterion is satisfied. It is a control-plane gate, not a prompt rule.

Component	What it is / does
Per-tenant theming	Design tokens + responsive breakpoints stored on the Project Profile flow verbatim into the manifest — one theme, no per-tenant code fork.
TestMgmt connector (Xray)	Authors test cases as Xray tests and imports execution results (idempotently). Headless, per-tenant, rate-governed.
Result import + defect loopback	A test run (Test Automation, 09b — the only phase that produces one) records a TestExecution + a test_report artifact; a **failing test opens a linked, traced defect** (a child WorkItem tied to the test and traced to the code); a later pass closes it.
Test-impact selection	Given a changed piece of code, walks the traceability graph to select just the tests that cover it.
Flaky quarantine	A test that flips pass/fail within one run is quarantined and excluded from the blocking gate (never silently blocks a merge, never silently passes).
Ephemeral preview env per PR	A companion-local seam: the app is brought up from the worktree, the URL is posted on the PR, and torn down on close. (Cloud preview is a Phase-5 drop-in behind the same seam.)
Migrated agents 05, 09, 09b, 10	UX Design (read-as-SoR + HTML wireframes), Testing, Test Automation (Playwright-MCP locator discovery + Page Object Model preserved), and Security (the nine review tracks + SBOM, with criticals opening defects).

Phase 3 — capstone slice (what this phase proves)

Design produced and gated through the WCAG checklist -> implementation reads the current design -> tests author to Xray and results import -> a failing test opens a traced defect that closes on a clean re-run -> a standalone security review -> a companion-local preview env. Verified end-to-end, with the full platform suite green across all phases.

2.5 Phase 4 — Brownfield-Full & Multi-Repo

Goal: the client front door. Onboard an *existing* project — any number of repositories — understand its code, **resume the lifecycle from the middle** instead of starting over, and keep cross-repo API contracts honest with **live drift detection**. This is the entry point most real clients will use.

Component	What it is / does
CodeIntel capability + built-in analyzer	A fixed 8-tool codeintel.* surface with a **built-in SOTA analyzer** (no license needed): git/Tornhill **hotspots** (churn × complexity), a dependency graph with **PageRank** + communities, dead-code and complexity, and decision archaeology. It runs in the **companion worktree** on a real clone; the platform stores only the index + artifacts, **never the source**. Repowise (or another engine) plugs in behind the same seam.
Knowledge Index	An RLS-aware chunk store with an always-on, **zero-cost keyword (BM25) baseline** plus an optional embedding seam (Voyage / Gemini). It feeds the manifest's retrieved-knowledge section and is **project-wide shared memory** across all of a project's repos.
Agent 00c — Intelligent Context Extractor	Migrated: turns a raw codebase into five intelligence layers (dependency graph, git analytics, dead code, decisions, health) registered as platform artifacts + indexed into the Knowledge Index. tech_lead gate.

Component	What it is / does
Agent 00 — Onboarding (full)	The brownfield front door: **discover** across the connected systems of record + code-intel, **register** what already exists (register-don't-recreate), **reconcile**, detect the **gaps**, and emit a **gap report** that assesses every lifecycle phase Complete/Partial/Missing and names the correct **resume phase**. product_owner gate.
Agent 00b -> API-Contract artifact	Cross-repo contracts as a **versioned api_contract registry artifact** (read-through, never a copied file): the provider repo exports it (semver-bumped); consumer repos read the current version at run time and record an implements edge. architect gate. (Agent 00d Snapshot is retired — continuity is now the database + durable runs.)
Multi-repo project model + per-repo manifest	One project owns **N repositories** (frontend/backend/service); topology reconciles to single/split/multi. The composed manifest is **scoped per repo** (that repo's artifacts + project-shared ones + the shared contract), while the registry, traceability, and Knowledge Index stay **project-wide shared memory**.
Cross-repo DAG edges	Dependencies that span repositories: implements traceability edges across repos + WorkItem.depends_on — the basis for cross-repo change-impact.
Live drift detection + drift gate	Two detectors: **contract drift** (an api_contract diverges from its consumers) and **pointer/hash drift** (a SoR artifact's external content changed since we indexed it). On detection the platform writes a **DriftRecord**, computes **change-impact via the traceability graph** (every downstream consumer/repo), and opens a decide_drift gate — architect for contract, tech_lead for pointer/hash. **Accept** re-indexes; **reject** flags for revert; nothing is silently written to a system of record. The signed Git push webhook is the live trigger (built on the Phase-2 reconcile).

Phase 4 — capstone slice (what this phase proves)

Onboard a **split FE/BE** project -> register-don't-recreate + the built-in code-intel running for real in a companion worktree -> a **gap report** with a resume phase -> the backend exports an api_contract, the frontend reads it read-through and records a cross-repo implements edge -> **resume mid-pipeline** triggers the next phase to its gate -> a **contract change opens a drift gate** whose change-impact **spans both repos**, decided by the architect. Verified end-to-end; the full platform suite is green across all phases.

2.6 Phase 4.5 — Local-Engine Execution & Interactive Clarifications

Goal: make “agents run on your own Claude Code” real end-to-end, and let an agent ask a genuine question mid-run instead of guessing — without ever billing while it waits.

Component	What it is / does
Companion-routed trigger	The REST trigger (and the UI) now resolves the acting user and routes the run to *their* enrolled companion (execution_mode=companion) instead of a server-side engine call — this is what makes Section 3's “run a phase” actually execute on your machine.
Interactive ask_human	An agent can open a pending **Clarification** mid-run and pause — the companion is idle and **no tokens are spent while it waits**. Answering (REST or the Clarifications screen) lets the companion pick the answer up on its next poll and resume in the same session.
Cost accounting fix	A run's cost is only set from the companion's final report — a paused clarification never shows a phantom cost.

Component	What it is / does
Offline -> human takeover	If the companion never reports (laptop closed, network down), the run parks the WorkItem for human takeover instead of hanging forever.

Phase 4.5 — capstone slice (what this phase proves)

Trigger a phase as yourself -> it routes to your companion -> the agent asks a question -> the run shows “needs answer” at **zero cost** while it waits -> you answer -> it resumes and completes -> gate -> publish. Plus: an offline companion parks the run for takeover instead of hanging.

2.7 Phase 4.7 — Web-App Productization: Auth & Admin (backend)

Goal: replace the pre-4.8 “paste a tenant id into a header” acceptance-slice shell with the real auth/admin backend a proper product needs — session login, self-serve signup, invitations, and role management — as the **design-invariant** contract the Phase 4.8 hi-fi screens are built against.

Component	What it is / does
Server-session auth	POST /auth/signup (atomically creates a new Tenant + its first admin User), POST /auth/login, GET /auth/me, POST /auth/logout — a real browser session (cookie), not a header you type in.
Invitations	Admin invites by email with initial project/role grants; the invitee accepts (sets their password) via a tokenised link; resend/revoke; no account enumeration on password reset.
User & role management	Users list, per-project role grants (Membership), gate-role coverage (flags e.g. “no reviewer -> merge gate unactionable”), the tenant-admin flag with a **last-admin guard** (can't demote the only admin) — both a tenant-admin *and* a project-admin can approve **any** gate on their scope.
Companion status + connect	The Connect-Machine flow (enrollment token, online/offline status) is now session-scoped to the logged-in developer instead of an ungated, tenant-wide list.
Tenant context from the session (hardened 2026-07-02)	How RLS learns which tenant a request belongs to was corrected after this Guide's own testing pass exercised it for the first time against the real (non-superuser) database role — see /docs/decisions/0002-rls-tenant-bootstrap-session.md for the full root cause and fix; the short version is in Section 4.4.

Phase 4.7 — capstone slice (what this phase proves)

From scratch, no shell step: signup -> admin invites a user -> user accepts + sets a password -> admin grants gate roles -> connect-machine status -> create a project -> trigger a phase as the authenticated user (routes to their companion) -> answer a clarification -> decide a gate -> preview the published artifact. This is exactly Section 5's Scenario 1, driven by the real REST surface underneath the 4.8 screens.

2.8 Phase 4.8 — Hi-Fi Web App (the UI in this guide)

Goal: build the real product UI — every journey (J1–J22) as a signed-off, hi-fi screen wired to live data — replacing the placeholder “Tenant + Project bar” shell that earlier guide revisions walked through.

Component	What it is / does
App shell + navigation	Login/Signup/AcceptInvite/ResetPassword (pre-auth) and, once signed in, a persistent sidebar (Dashboard, Work Items, Phases & Runs, Gates, Clarifications, Artifacts & Traceability, plus Project and Admin sections) with a tenant/project switcher — this is what Section 3.2 walks through screen by screen.
Work Items board + detail	Create/list/filter work items (Epic/Task/Defect), a type-adaptive lifecycle rail per item, in-place edit, and a task-breakdown fan-out from an approved epic into build-tier tasks.
Phases & Runs	Pick a work item, trigger any phase for it (blocked with a link to Connect Machine if your companion is offline), and watch the run timeline live.
Gates, Clarifications, Criteria-gate	Real approve/reject with the required role enforced, a generalized checklist-condition gate (e.g. the WCAG gate), and a live answer form for pending questions.
Artifacts, Traceability, Publish	Registry + preview + version history, a lineage graph, and a gate-respecting publish action to the live system of record.
Onboarding (brownfield front door)	Upload-and-extract source documents behind an adopt gate, register existing Confluence/Jira (no duplicates), a gap report with a resume-phase recommendation. **Code-intelligence indexing is the one still-open piece** — see the callout below.
Setup: Integrations, Repositories, Connect Machine	Attach Confluence/Jira/GitHub/Figma/Xray from a form (no shell script), manage the project's repositories, and generate a companion enrollment token with the exact two terminal commands to run.
Admin: Users & Invitations, Roles & Memberships	Invite teammates (multiple project/role grants per invite), see pending invitations alongside active members, and grant/revoke per-project roles or the tenant-admin flag from a matrix.

What's genuinely still not UI-driven — brownfield code-intelligence dispatch

Every brownfield onboarding step above works from the browser **except** triggering the built-in code-intelligence analyzer itself: the backend endpoint only *records* an already-computed result, and there is deliberately no live command channel yet from the platform to a developer's companion to run it on demand (a **documented, deferred architecture decision**, not an oversight — Phase 4.7's own spec calls it out). Because of this, Section 5's brownfield scenarios (5.4/5.5) are left as they were in the previous guide revision — API/shell-annotated — rather than rewritten to a UI flow that doesn't fully exist yet. Everything else in this guide, greenfield and brownfield alike, is UI-first.

Phase 4.8 — capstone slice (what this phase proves)

Every screen renders every labelled state from the hi-fi design gallery with visual parity, wired to live endpoints — not mock data. Section 3.2 and Section 5's greenfield scenarios are a straight-through click-path proof of this.

3. The Local Execution & Testing Guide

This is a hands-on guide for a developer to run the platform locally and *see* every workflow work — in the browser, the API, the Temporal dashboard, and the object store. It is not about unit tests. Follow Part A to get the core running with zero external accounts; Part C adds the optional live connectors.

Before you start

Complete the prerequisites in **Section 4** first (Docker Desktop running, .env created). **Agents run on your own Claude Code** (req #1): every phase — planning, build, and review — executes on the machine running the **companion** daemon, using your **logged-in local claude** (your subscription), **not** an API key. So instead of setting ANTHROPIC_API_KEY, you log in once (claude / claude setup-token) and run the companion (Step 4). A server-side ANTHROPIC_API_KEY path still exists as a headless/CI fallback only. On Windows without make, use ./make.ps1 <target>.

3.1 Part A — Bring up the platform (no external accounts needed)

Step 1 — Start the infrastructure

```
# from the repo root
docker compose up -d        # postgres (:5433), temporal (:7233 + UI :8080), minio (:9000 + console :900
docker compose ps              # all services should be 'running' / healthy
```

What you should see: open the **Temporal UI** at http://localhost:8080 (empty workflow list for now) and the **MinIO console** at http://localhost:9001 (login minioadmin / minioadmin; the sdlc-artifacts bucket already exists).

Step 2 — Migrate the database and seed the demo tenant

```
make migrate     # creates every table + its RLS policy (runs as the DB owner)
make seed        # seeds a demo tenant, project, profile, agent specs, connectors
```

What you should see: seed prints the created tenant/project ids and a connector count. (If make migrate reports an ownership error on an ALTER, run it as the postgres superuser — see Section 4.4.)

Step 3 — Run the application (API + worker + frontend)

```
make dev         # = docker compose --profile app up
#   backend API  -> http://localhost:8000
#   React UI     -> http://localhost:5173
#   worker       -> connects to Temporal (durable runs execute here)
```

What you should see: the React app loads at http://localhost:5173; the worker logs “started” and registers with Temporal. A quick API smoke check:

```
curl http://localhost:8000/api/projects/ -H "X-Tenant-ID: <demo-tenant-id>"
# -> JSON list including the seeded demo project
```

Step 4 — Run your local companion (so phases execute on YOUR Claude Code)

Every phase the platform dispatches runs on the machine running the companion, against your logged-in claude — no API key. Log in once, enroll this machine for your user, then leave the daemon running:

```
# 1) log Claude Code in once (interactive, or a long-lived token for headless):
claude            # sign in to your subscription   (or: claude setup-token)

# 2) enroll this machine for YOUR user (prints a one-time token):
python -m runner.companion enroll --api-url http://localhost:8000 \
    --tenant $TENANT --project $PROJECT --label my-laptop --developer-id $USER

# 3) run the headless daemon — it claims your dispatched phases and runs them locally:
python -m runner.companion serve --api-url http://localhost:8000 \
    --tenant $TENANT --token <enrollment-token>   # add --repo-path/--worktrees for builds
```

What you should see: the daemon logs that it preflighted claude on PATH (and dropped any ANTHROPIC_API_KEY so your subscription is used), then idles polling for work. When you trigger a phase (next), it claims and runs it; an agent that needs input surfaces a question on the **Clarifications** tab and waits at no cost.

3.2 Part B — Drive the planning slice in the browser (real Phase 4.8 screens)

This is the core, no-external-accounts workflow, entirely in the browser — the only exceptions anywhere in this walkthrough are the two local terminal commands in Step 4 above (claude login and running the companion). Open http://localhost:5173: you land on **Login**. This guide builds **your own** tenant from scratch rather than the seeded demo — click **"Create a workspace"** to reach **Signup**.

> **Cost & interaction note**
>
> Phases run on your **local Claude Code subscription** via the companion — there is no per-call API charge and no ANTHROPIC_API_KEY needed. If an agent needs information it cannot find, it asks on the **Clarifications** screen and **pauses at zero cost** until you answer (the model is idle while it waits) — then continues in the same session. To see the whole loop quickly, trigger a single phase (Ideation).

Screen	What to do	What you should see (the proof)
Signup (/signup)	Full name, work email, a password, and your workspace/org name; **Create workspace**.	You land signed in as the tenant's first admin — no shell step created you. A confirmation screen hands you to the Setup Checklist.
Setup Checklist (/setup)	Follow its steps in order: create a project, connect your machine, invite teammates (optional for a solo test), connect integrations (optional — Part C).	Each step ticks off as you complete it.
Create Project (/projects/new)	Name it, pick Greenfield, Save.	You land on the new project's Project Profile, and it becomes your selected project everywhere (top-bar switcher) — as the creator you're automatically granted the project-admin role, so you can approve every gate on it yourself.
Connect Machine (/connect)	Pick this project, **Generate enrollment token**.	A one-time token is shown once, with the exact two commands to paste into your terminal — the same claude + companion serve pair from Step 4, now pre-filled with your token. Once the daemon checks in, this screen (and the sidebar) shows **Connected**.

Screen	What to do	What you should see (the proof)
Project Profile (/profile)	Fill description, tech stack, important context; Save.	A new profile version is saved; it becomes the agent's context next run.
Work Items (/work-items)	**+ New Work Item** -> a title, type **Epic** -> Create.	You land on the new epic's detail page with a full 16-phase lifecycle rail, all Upcoming.
Phases & Runs (/runs)	Pick your work item, click **Ideation** to trigger it.	A Run appears and moves running -> awaiting_gate as your companion executes it. In the Temporal UI (:8080) a new AgentRunWorkflow is visible. A **"needs answer"** badge appears on the Clarifications nav item if the agent asks a question.
Clarifications (/clarifications)	If the badge is showing, open this screen, answer the question(s), Send.	The agent resumes in the same session (it was paused at zero cost); the badge clears.
Gates (/gates)	Find the pending approval; Approve (you hold the role — project-admin can approve any gate).	The run resumes to completed; the workflow finishes in the Temporal UI.
Artifacts & Traceability (/artifacts, /traceability)	Open the ideation artifact; open the lineage view.	The artifact's status flips draft -> published (body in MinIO under drafts/ and snapshots/); the traceability graph shows it and its derives/tests/... edges — once external SoRs are connected, nodes deep-link to the live Confluence/Jira items.

Repeat the trigger for **Requirements -> PRD -> Architecture** from Phases & Runs to walk the full planning chain. Each run reads the artifacts the previous phase published (proving the platform — not a local file — is the source of truth), and each is independently gated. This exact walkthrough, scripted start to finish, is Section 5's Scenario 1.

3.3 Part C — Optional: exercise the live connectors

These steps need real (test) cloud accounts. Each is independent — do only the ones you have credentials for. Set the relevant variables in .env (Section 4) and make seed again so the demo tenant gets the connector, then restart make dev.

Atlassian — Confluence + Jira (Phase 1)

- Set ATLASSIAN_*, CONFLUENCE_SPACE_KEY, JIRA_PROJECT_KEY.
- **Greenfield:** approve the PRD gate -> the PRD **publishes as a Confluence page**; approve the backlog -> epics/stories **appear in Jira**. The Traceability tab now deep-links to those live items.
- **Brownfield:** POST a Word/Excel document to /api/source-documents/ -> it is extracted into reviewable internal artifacts (gated before adoption); existing Jira issues are registered without duplicates.

GitHub — the build path (Phase 2)

Set GITHUB_API_URL, GITHUB_TOKEN (a PAT with repo scope), GITHUB_REPO ("owner/name"), GITHUB_DEFAULT_BASE, GITHUB_WEBHOOK_SECRET. Then enroll a companion on your machine and run a build:

```
# 1) enroll this machine as a runner (prints a one-time token)
python -m runner.companion enroll --api-url http://localhost:8000 \
    --tenant <tenant-id> --project <project-id> --label my-laptop --repos owner/name

# 2) run an assigned build task in a managed worktree
python -m runner.companion run --task task.json \
    --repo-path ~/repos/your-clone --worktrees ~/sdlc
```

What you should see: a feature/<key> branch is created in a worktree, tests run locally, a **draft PR opens on GitHub**, code review runs, and the Temporal UI shows the BuildWorkflow parked at the **reviewer merge gate**. Approve it (Gate Approvals) and the PR **merges**. Push a manual commit to the branch and the signed webhook reconciles it (the artifact's sha pointer moves; no duplicate).

Figma + Xray (Phase 3)

- **Figma:** set FIGMA_TOKEN + FIGMA_FILE_KEY (a personal-access token + a test file). The design phase reads the *current* design tokens/components from that file (read-as-SoR) instead of inventing them.
- **Xray:** set XRAY_CLIENT_ID + XRAY_CLIENT_SECRET + XRAY_PROJECT_KEY. The test phase authors tests into Xray and imports results; a failing import opens a defect WorkItem linked to the test and traced to the code, which closes automatically on a clean re-run.

3.4 Part D — See the cross-cutting guarantees with your own eyes

Guarantee	How to observe it
Durability	In the Temporal UI (:8080), open a running workflow — it shows each activity, retries, and the gate wait. Restart the worker container mid-run; the workflow resumes where it left off.
Gate-then-publish	On the Artifacts tab, an artifact stays draft until you approve its gate — then it becomes published. There is no way to publish without an approval.
Tenant isolation (RLS)	Create a second tenant and call the API with its X-Tenant-ID: it sees none of the first tenant's projects, runs, artifacts, or defects.
Provenance / cost	The Run Timeline shows each run's engine, model, cost (USD), and Temporal workflow id; the Event ledger records who approved what and when.
Artifact bodies in S3	The MinIO console (:9001) shows artifact draft bodies and immutable post-approval snapshots — the registry holds only pointers + hashes.

> **"Everything works" checklist**
>
> (1) infra healthy; (2) migrate + seed succeed; (3) UI loads and the tabs work; (4) a planning phase runs to completed through its gate and publishes an artifact; (5) traceability shows the lineage; (6) the Temporal UI shows the durable workflow; (7) MinIO shows the artifact body; (8) optional: a live PRD reaches Confluence, a backlog reaches Jira, a build reaches a GitHub PR, and a failing test opens a linked defect.

4. Setup, Credentials & Prerequisites

Everything a developer must install, configure, or obtain before running Section 3. Copy .env.example to .env and fill the values below.

4.1 Prerequisites to install

Tool	Why	Notes
Docker Desktop	Runs the whole stack (Postgres, Temporal, MinIO, API, worker, frontend).	Must be **running** before docker compose up. Windows: WSL2 backend recommended.
Python 3.12 + a repo-root .venv	Local tooling: fast pytest/ruff, the companion runner CLI.	python -m venv .venv then install backend/requirements-dev.txt. Tooling only — the app runs in Docker.
Git	Version control + the companion's managed worktrees.	Any recent version.
make (optional)	Convenience targets.	Windows without make: use ./make.ps1 <target>.

4.2 Environment variables (.env)

Required for the core local run

Variable	Purpose	Default / value
DJANGO_SECRET_KEY	Django signing key (dev).	any dev string
POSTGRES_* / DATABASE_URL	App database + the non-superuser RLS role.	preset in .env.example (app uses sdlc_app; host port 5433)
TEMPORAL_HOST / NAMESPACE / TASK_QUEUE	Durable workflow engine.	temporal:7233 / default / sdlc-planning
CORS_ALLOWED_ORIGINS	Lets the React app call the API.	http://localhost:5173
S3_* (MinIO)	Object store for artifact bodies + snapshots.	minioadmin / minioadmin, bucket sdlc-artifacts
ANTHROPIC_API_KEY	**Required to actually run agents** (the ClaudeDriver). Each run costs a little.	sk-ant-... from console.anthropic.com
DEFAULT_PLANNING_MODEL	Which Claude model planning runs use.	claude-opus-4-8 (use a cheaper model to reduce cost)

Optional — only for the live connector walkthroughs (Part C)

Variable group	Enables	Where to get it
ATLASSIAN_SITE_URL, ATLASSIAN_EMAIL, ATLASSIAN_API_TOKEN, ATLASSIAN_CLOUD_ID, CONFLUENCE_SPACE_KEY, JIRA_PROJECT_KEY	Phase 1: PRD -> Confluence, backlog -> Jira, traceability deep-links, brownfield onboarding.	API token: id.atlassian.com -> Security -> API tokens. Use a TEST space/project.
GITHUB_API_URL, GITHUB_TOKEN, GITHUB_REPO, GITHUB_DEFAULT_BASE, GITHUB_WEBHOOK_SECRET	Phase 2: companion build -> draft PR -> review -> merge gate -> merge; webhook reconcile.	A PAT with repo scope (github.com -> Settings -> Developer settings). Add these keys to .env (not in the template by default).

Variable group	Enables	Where to get it
FIGMA_TOKEN, FIGMA_FILE_KEY	Phase 3: read the current design as the source of truth.	Figma -> Settings -> Personal access tokens; a test file's key from its URL.
XRAY_CLIENT_ID, XRAY_CLIENT_SECRET, XRAY_PROJECT_KEY	Phase 3: author tests + import results; failing test -> linked defect.	Xray Cloud -> Global Settings -> API Keys. Use a TEST project.

Security

Real credentials live only in .env, which is git-ignored — never commit them. Always use **test** spaces/projects/repos, not production ones. Without any connector creds the core local run (Parts A & B) still works fully on the internal source of truth.

4.3 Ports reference

Service	URL	Login
React UI (Experience plane)	http://localhost:5173	—
Backend API	http://localhost:8000/api	X-Tenant-ID header
Temporal Web UI	http://localhost:8080	—
MinIO console (object store)	http://localhost:9001	minioadmin / minioadmin
Postgres (host-mapped)	localhost:5433	postgres / postgres (admin)

4.4 Manual setup notes & gotchas

- **Migrate as the DB owner.** The app connects as the non-superuser sdlc_app role (so RLS applies). Schema changes (ALTER) must run as the postgres superuser. If make migrate fails on an ALTER, run migrations with the superuser DSN: DATABASE_URL=postgres://postgres:postgres@localhost:5433/sdlc_platform.
- **Postgres is on host port 5433**, not 5432, to avoid clashing with a native Postgres. In-container services still use postgres:5432.
- **If the stack drops** (a “Socket is not connected” error on :5433 usually means Docker Desktop stopped), start Docker Desktop and run docker compose up -d again.
- **Windows without make:** use ./make.ps1 <target> (e.g. ./make.ps1 dev).
- **First frontend start** runs npm install inside the container — the very first make dev takes a few minutes while node modules download.
- **The companion runner** runs on the developer's own machine (not in Docker) via python -m runner.companion ...; it connects outbound to the platform API.

- **Login/signup returning 401 or “Invalid email or password” with correct credentials** on an older local database means migration apps/tenancy/migrations/0005_user_rls_auth_bootstrap.py hasn't run yet — make migrate (as the postgres superuser, same as above) applies it. See /docs/decisions/0002-rls-tenant-bootstrap-session.md for why this exists: resolving who's logged in has to work *before* the tenant is known, which the original RLS policy on the users table didn't allow for.

> **Quick start, condensed**
> cp .env.example .env (set ANTHROPIC_API_KEY) -> docker compose up -d -> make migrate -> make seed -> make dev -> open http://localhost:5173 and run a phase through its gate.

This document describes Phases 0–4.8 (complete and tested). Section 5 is an explicit test-through inventory of what works today. The next milestone is Phase 5 — Enterprise Hardening & Cloud Migration (RBAC + segregation of duties, secrets vault, prompt-injection hardening, immutable audit/compliance, the eval harness, and the cloud runner fleet).

5. End-to-End Testing — Scenario Playbook

This section is a hands-on playbook for an engineer to test the platform across the scenarios that matter, **each built from scratch** — you create your own tenant, user, project and companion; nothing here uses the seeded demo tenant. Do **Section 3** bring-up once, then the shared **5.1 preflight**, then run whichever scenarios you need. Scenarios 1, 2, and 5 (greenfield + cross-cutting) are **UI-only** except the two irreducible local commands (claude login, running the companion — Section 3.1 Step 4). Scenarios 3 and 4 (brownfield) still carry API/shell steps for the one piece that isn't wired to the browser yet — code-intelligence indexing — a deliberate, documented scope decision explained in 2.8; those two scenarios are otherwise unchanged from the previous guide revision pending that follow-up.

> **Two ground rules for every scenario**
>
> (1) **From scratch:** you create the tenant/user/project yourself, via the real Signup screen — do not use the seed demo tenant. (2) **Agents run on your Claude Code:** every phase executes on your machine via the **companion** daemon using your logged-in claude (no API key); you only ever click and type in the web app. The handful of remaining API calls (brownfield code-intel only) carry -H "X-Tenant-ID: $TENANT".

5.1 Preflight — build your own world (shared by all scenarios)

Step 1 (infra). Bring the stack up once (Section 3.1): docker compose up -d -> make migrate -> make dev. You do **not** run make seed — you are building everything yourself.

Step 2 (UI) — sign up. Open http://localhost:5173/signup: name, work email, password, workspace name -> **Create workspace**. This atomically creates your tenant and you as its first admin (and seeds that tenant's agent specs) — no curl, no shell. You're signed in immediately. As the tenant's sole admin, and as the project-admin you're automatically granted on any project you create (5.2's first step), you can approve **every** gate yourself — invite a teammate only if you specifically want to test role-scoped gating.

Step 3 (UI, optional) — invite a teammate + grant roles. Only needed if you want more than one person: **Users & Invitations** (sidebar -> Workspace Admin) -> **+ Invite** -> email + one or more project/role grants -> Send invite. The invite shows as **Invited · pending** with Resend/Revoke until accepted. To change anyone's roles later, **Roles & Memberships** is a live grant/revoke matrix per project, plus the tenant-admin toggle (guarded — you can't demote the last admin).

Step 4 (CLI, irreducible) — log in Claude Code and run your companion. This is what makes phases run on your machine. In the UI, open **Connect Machine** (sidebar), pick your project, and click **Generate enrollment token** — it shows the token once, with the exact two commands below already filled in for you to paste into your terminal:

```
claude            # sign in to your subscription   (or: claude setup-token for headless)
python -m runner.companion serve --api-url http://localhost:8000 \
    --tenant $TENANT --token <enrollment-token>      # add --repo-path/--worktrees for builds
```

Once the daemon checks in, Connect Machine (and the sidebar) shows **Connected**.

How a gate or a question is handled (the same in every scenario)

When a run reaches a human gate it appears on the **Gates** screen — Approve (project-admin/tenant-admin can approve any gate; otherwise you must hold that gate's role). If an agent needs information it posts a question on the **Clarifications** screen and the run shows a **“needs answer”** badge; answer it and the run continues at no extra cost.

5.2 Scenario 1 — Greenfield, NO external integrations

Prove the full **planning/doc lifecycle** with everything kept internal — no Confluence/Jira/Figma/Xray. Artifacts publish to the platform's internal store (bodies in MinIO). Every step below is a click in the browser.

- **Create Project** (/projects/new): name it, Track **Greenfield**, Save. You're auto-granted project-admin on it (5.1 Step 2), and it becomes your selected project. Then **Connect Machine** (5.1 Step 4) so phases have somewhere to run.
- **Project Profile** (/profile): fill description, tech stack, important context; Save.
- **Work Items** (/work-items): **+ New Work Item** -> title, type Epic -> Create. You land on its detail page with the full lifecycle rail.
- **Phases & Runs** (/runs): select the work item, trigger phases in order: **Ideation -> Requirements -> PRD -> Architecture** (and 05/06 if you wish). Each runs on your companion.
- **Clarifications** (/clarifications): if a run shows “needs answer”, answer it -> the run resumes in the same session.
- **Gates** (/gates): approve each phase's gate -> the run completes and its artifact flips draft -> published.
- **Artifacts & Traceability** (/artifacts, /traceability): see each artifact (body in the MinIO console :9001) and the lineage graph (derives edges) — all sor=internal.

Scenario 1 proof

A greenfield project went idea -> requirements -> PRD -> architecture entirely on your local Claude Code, each phase human-gated, every artifact published internally with full lineage — **zero external accounts**, and every step above was a click, not a shell command. (The build/merge + live-SoR phases are Scenario 2.)

5.3 Scenario 2 — Greenfield, WITH external integrations

Same lifecycle as Scenario 1, but real systems of record: PRD -> Confluence, backlog -> Jira, UX from live Figma, build -> a GitHub PR, tests -> Xray. Put the credentials in .env first (Section 4.2 — the connect form below still needs somewhere to source tokens/URLs from), then **attach each connector to YOUR project from the browser**:

- **Integrations** (sidebar, under Project): a card per capability (Docs, Task Tracker, Code Host, Design Source, Test Mgmt), each showing **Not connected**.
- Click a card -> **Connect a capability** (/integrations/<capability>): pick the vendor, fill its binding fields (e.g. Confluence's space key, Jira's project key, GitHub's repo) and the secret (API token/PAT), **Test connection**, then Save.

- Repeat per capability you want live: Docs -> Confluence, Task Tracker -> Jira, Code Host -> GitHub, Design Source -> Figma, Test Mgmt -> Xray. The Integrations grid now shows each as **Connected**.

Now run the lifecycle exactly as Scenario 1 (Phases & Runs + Clarifications + Gates). The same two moves per phase — **trigger**, then **approve** — now also write to the live tools. Per-phase agent keys, gate roles, and the live proof:

Phase — agent_key	Gate role	Live proof on approval
PRD — 03_prd	product_owner	A **Confluence page** is created; the artifact's external_ref.url + the Traceability node deep-link to it.
UX Design — 05_ux_design	designer	Reads your **live Figma** tokens/components; the **WCAG gate** blocks approval until every required criterion is ticked.
Task Breakdown — 06_task_breakdown	tech_lead	**Epics/stories appear in Jira**; child WorkItems + story->requirement trace links are created.
Implementation — 07_implementation	tech_lead	Runs on your companion worktree -> a **draft PR opens on GitHub** (start the companion with --repo-path/--worktrees).
Code Review — 08_code_review	reviewer	Runs on your companion; findings become child WorkItems; the 07<->08 loop re-runs until clean.
Testing — 09_testing	qa_lead	**Authors the test plan and test cases** (planning only; a reviewer exports accepted cases to Xray). **Test Automation (09b)** then runs the tests, imports results to Xray, and a failing test opens a linked, traced defect that closes on a clean re-run.
Merge	reviewer	Approve the merge gate -> the **GitHub PR merges**; the WorkItem moves to done.

Scenario 2 proof

The same greenfield run now produced a **Confluence PRD**, a **Jira backlog**, **UX from live Figma** behind a WCAG gate, a **merged GitHub PR**, and **Xray tests with a linked defect** — each gated, each on your local Claude Code.

5.4 Scenario 3 — Brownfield, uploaded resources only (no external tools)

Onboard an existing project using only what you **upload** — no Confluence/Jira. The platform assesses what exists and tells you which phase to resume. Brownfield onboarding services have no dedicated UI yet, so the onboarding steps are API/shell; gates are approved in the UI. **Test onboarding at different stages** by varying what you provide:

- **(API) Create the brownfield project** + grant roles + enroll/serve the companion (5.1): curl -s $H -X POST $API/projects/ -d '{"name":"Legacy Shop","track":"brownfield"}'.
- **(API) Upload static resources:** curl -s -H "X-Tenant-ID:$TENANT" -F file=@./old-spec.docx $API/projects/$PROJECT/source-documents/ (repeat for .xlsx/.pdf). Each is extracted into reviewable draft artifacts.
- **(UI) Adopt them behind a gate:** approve adoption — or POST /source-documents/{id}/adopt/ {"user_id":"$USER"}.

- **(shell) Optional code intelligence** on a local clone (built-in analyzer; the platform stores the analysis + index, never your source): run BuiltinCodeIntel().index(workspace=...) + tools.register_artifact(...) (see the snippet in 5.5).
- **(shell) Gap report -> resume phase:** build_gap_report(tenant_id=$TENANT,project_id=$PROJECT) returns resume_phase + the blocking gaps.
- **(UI) Resume mid-pipeline:** trigger the phase it named (e.g. 04_architecture) on a WorkItem and approve its gate — onboarding hands straight into the normal pipeline.

> **Onboarding at different stages — vary the inputs**
>
> **Empty** (upload nothing) -> the gap report resumes at ideation. **Partial** (upload a requirements doc only) -> it resumes at prd/architecture. **Mature** (docs + code intelligence) -> it assesses implementation/testing as present and resumes later. Run the gap report after each upload to watch the resume point move.

5.5 Scenario 4 — Brownfield, WITH external tools

As Scenario 3, plus live tools: register the project's **existing** Confluence/Jira (register-don't-recreate — no duplicates), and exercise the multi-repo API-contract + live drift gate. Attach connectors to your tenant (5.3 snippet) first.

```
# 1) (shell) register repos (multi-repo) + (API) register existing Confluence/Jira
docker compose exec backend python manage.py shell -c "\
from apps.projects.repos import add_repository as A;\
A(tenant_id='$TENANT',project_id='$PROJECT',name='owner/web',role='frontend');\
A(tenant_id='$TENANT',project_id='$PROJECT',name='owner/api',role='backend')"
curl -s $H -X POST $API/projects/$PROJECT/onboard-atlassian/   # existing items, deduped

# 2) (shell) code intel on a clone -> gap report -> versioned API contract (read-through)
docker compose exec backend python manage.py shell -c "\
from integrations.onboarding import build_gap_report;\
from integrations.api_contract import export_api_contract, read_api_contract;\
print('RESUME ->', build_gap_report(tenant_id='$TENANT',project_id='$PROJECT')['resume_phase']);\
c=export_api_contract(tenant_id='$TENANT',project_id='$PROJECT',body='GET /users',source_commit='abc123'
```

```
# 3) (shell) breaking contract change -> drift gate with cross-repo impact -> architect decides
docker compose exec backend python manage.py shell -c "\
from integrations.api_contract import export_api_contract;\
from integrations.drift import detect_contract_drift, decide_drift;\
b=export_api_contract(tenant_id='$TENANT',project_id='$PROJECT',body='GET /users (v2)',change='major');\
d=detect_contract_drift(tenant_id='$TENANT',contract_artifact_id=b['artifact_id'],new_semver='2.0.0');\
print('impacted ->', d['impacted']);\
print(decide_drift(tenant_id='$TENANT',drift_id=d['drift_id'],user_id='$USER',decision='accept'))"
#   live alternative: push to the provider repo so the signed GitHub webhook opens the drift gate
```

(UI) Resume mid-pipeline at the gap-report phase and continue exactly as Scenario 2 (phases run on your companion; PRD/backlog/PR/tests hit the live tools).

> **Scenario 4 proof**
> An existing multi-repo project was onboarded with its real Confluence/Jira registered (no duplicates) and its code indexed, produced a **gap report + resume phase**, exported a **versioned API contract read across repos**, and a **breaking change opened a drift gate** whose impact spanned both repos.

5.6 Scenario 5 — Interactive clarifications & offline takeover (cross-cutting)

These prove the two Phase-4.5 guarantees directly and apply on top of any scenario above. Everything except stopping/restarting the companion itself is a click.

- **Zero-cost clarification (UI):** trigger any phase; if the agent asks, the Clarifications nav item shows a **“needs answer”** badge and the run sits idle (your claude is not generating, so nothing is billed). Open **Clarifications**, answer -> the run resumes in the same session.
- **Durability (no re-ask):** stop the companion (Ctrl-C) while a clarification is answered but the phase hasn't reported, then re-trigger the phase from **Phases & Runs** — the agent reuses your earlier answer (folded into its manifest) instead of asking again.
- **Offline takeover:** stop the companion, trigger a phase, and leave it; after the report timeout the run parks the WorkItem for **human takeover** (visible on the **Offline Takeover** screen — Run status failed + a takeover flag) rather than hanging — restart the companion, or take it over by hand, to resume.
- **Local engine, not the API (verify):** the companion log shows it dropped ANTHROPIC_API_KEY and used claude on PATH; usage shows on your Claude subscription, not an API bill. Connect Machine shows **Connected** while it's up.

5.7 If something stalls

Almost always it is the environment, not the platform. Quick checks:

- **Infra down:** a “Socket is not connected” on :5433 means Docker Desktop stopped — start it and docker compose up -d again.
- **Migrations:** if make migrate errors on an ALTER, run it with the postgres superuser DSN (Section 4.4).
- **Login/signup fails with correct credentials:** a stale local database is missing the RLS-auth-bootstrap migration — see the Section 4.4 gotcha and ADR 0002.
- **Phase never runs / stays pending:** your **companion isn't running** or isn't enrolled for this project — check **Connect Machine** shows **Connected**, that serve is up in your terminal, and that claude is logged in on that machine.
- **Live SoR step does nothing:** the connector isn't attached to *your* project — check **Integrations** shows it **Connected** (5.3) — or the creds aren't in .env; restart make dev after editing .env.
- **Gate won't approve (403):** you don't hold that gate's role on the project and aren't a project-/tenant-admin — grant the role via **Roles & Memberships** (5.1 Step 3).

- **Full automated proof anytime:** cd backend && pytest -q — including tests/test_acceptance_phase47.py (the full from-scratch REST spine: signup -> invite -> roles -> connect -> trigger -> clarify -> gate -> preview) and tests/test_acceptance_phase4.py (brownfield/multi-repo).